Problem 6

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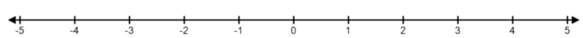
Problem 1 *Consider an object moving along a line with velocity* $v(t) = \pi \sin(\pi t)$ *on* $[0, 2]$ *and initial position* $s(0) = 0$. *Time is measured in seconds and velocity in m/s.*

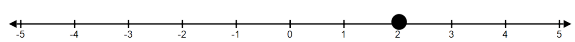
(a) *Determine the position function,* $s(t)$, *on* $[0, 2]$.

Explanation. General antiderivative of $v(t)$: $s(t) = -\cos(\pi t) + C$
Apply initial position: $s(t) = -\cos(\pi t) + C \implies 0 = s(0) = -\cos(0) + C \implies C = 1$
Position function: $s(t) = -\cos(\pi t) + 1$

(b) *Mark the position of the object at the time* $t = 1$ *on the line below.*



Explanation. $s(1) = -\cos(\pi) + 1 = -(-1) + 1 = 2$



(c) *Determine the average velocity,* v_{av}, *of the object during the interval* $[0, 2]$.

Explanation.

$$\begin{aligned} v_{av} &= \frac{s(2) - s(0)}{2 - 0} \\ &= \frac{-\cos(2\pi) + 1 - (-\cos(0) + 1)}{2} \\ &= \frac{-\cos(2\pi) + \cos(0)}{2} \\ &= \frac{-1 + 1}{2} = 0 \end{aligned}$$

(d) *Determine when the motion is in the positive direction.*

Explanation. Motion is in the positive direction when $v(t) > 0$.

$$\begin{aligned} v(t) > 0 &\iff \pi \sin(\pi t) > 0 \\ &\iff \sin(\pi t) > 0 \\ &\implies 0 < \pi t < \pi \\ &\iff 0 < t < 1 \end{aligned}$$

(e) *At what time (or times) is the object farthest from the origin?*

Explanation. Object is farthest from the origin when $\mid s(t) - 0 \mid = \mid 1 - \cos(\pi t) \mid$ is maximized. Since $s(t) \geq t$ for t in $[0, 2]$ we need to maximize $s(t)$. Finding critical points on the open interval $(0, 2)$.

$$\begin{aligned} v(t) = 0 &\iff \pi \sin(\pi t) = 0 \\ &\implies \pi t = \pi \\ &\iff t = 1 \end{aligned}$$

Finding the absolute maximum, we have to check the endpoints and the critical points:

$$\begin{aligned} s(0) &= 0 \\ s(1) &= 2 \\ s(2) &= 0 \end{aligned}$$

Hence the object is farthest from the origin at $t = 1$.